

Inference with mixed effects

Agenda

- + Project part 3 due last day of class
 - + No additional analysis needed; just combining work from first two parts and adding introduction, conclusion, etc.
 - + There will be time during the last week of class to work on the project
- + Exam 2 Friday
 - + Review day on Wednesday in class

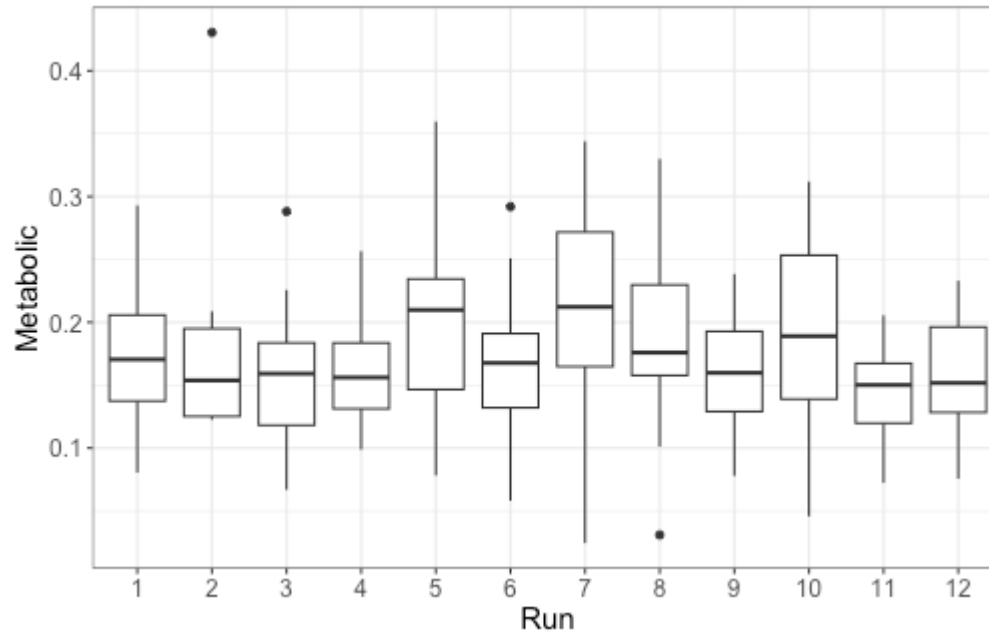
Data and Goal

We have data on 197 early-stage *Bugula neritina*, with information on

- + Run: which repetition of the experiment the individual was recorded in
- + Mass: Mass of the individual (in micrograms)
- + Metabolic: Recorded metabolic rate (rate of energy consumption) of the individual (in mJ per hour)

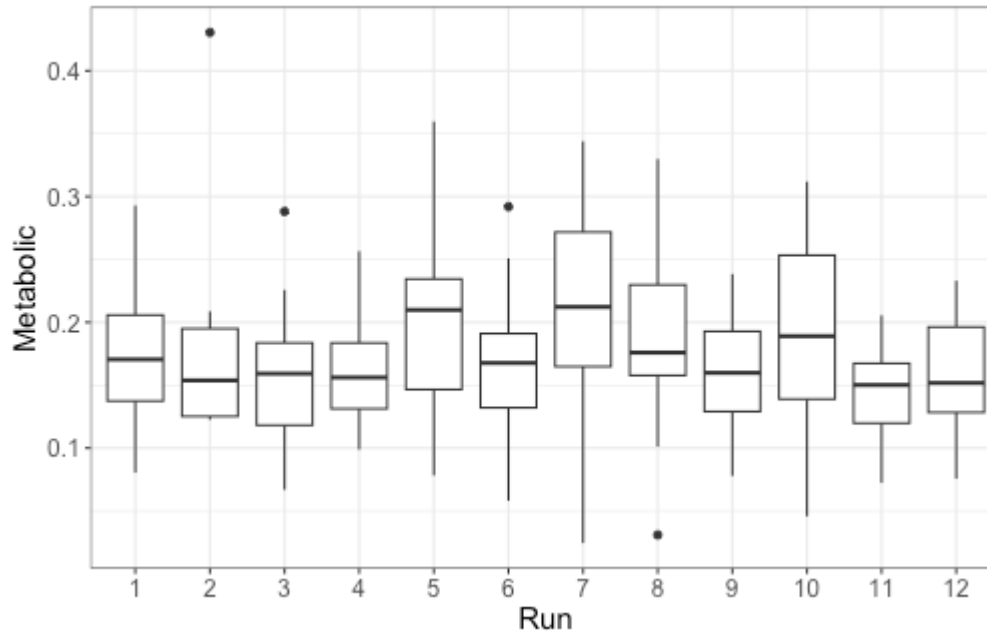
Goal for this class: Is there systematic variation between different runs (i.e., is there any correlation due to Run)?

Visualizing runs



Does it look like there is variation from run to run (i.e., do we need u_i)?

Visualizing runs



It is hard to tell if there are actually differences between runs, or if observed differences are just due to chance.

How can I estimate variability between runs?

Initial random intercepts model

variability between runs
variability within a run

$$Metabolic_{ij} = \underbrace{\beta_0 + u_i}_{\text{random intercept for each run}} + \varepsilon_{ij} \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2), \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

```
m1 <- lmer(Metabolic ~ (1|Run), data = bugula_early)
summary(m1)
```

Random effects:

Groups	Name	Variance	Std.Dev.
Run	(Intercept)	0.000131	0.01145
Residual		0.004181	0.06466

$$\hat{\sigma}_u^2 = 0.000131$$

$$\hat{\sigma}_\varepsilon^2 = 0.0042$$

$$p_{\text{group}} = \frac{0.000131}{0.000131 + 0.0042} \approx 0.03$$

About 3% of variation in metabolic rate is explained by differences in runs

Initial random intercepts model

$$\text{Metabolic}_{ij} = \beta_0 + u_i + \varepsilon_{ij} \quad u_i \stackrel{iid}{\sim} N(0, \sigma_u^2), \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

$$\hat{\beta}_0 = 0.175, \quad \hat{\sigma}_u^2 = 0.00013, \quad \hat{\sigma}_\varepsilon^2 = 0.0042$$

How would we test whether there is systematic variation between runs?

$$H_0: \sigma_u^2 = 0$$

$$H_A: \sigma_u^2 > 0$$

$$\Rightarrow \rho_{\text{group}} = 0$$

$$\Rightarrow \rho_{\text{group}} > 0$$

Potential test statistic:

$$\hat{\rho}_{\text{group}} = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \hat{\sigma}_\varepsilon^2}$$

we need null distribution of $\hat{\rho}_{\text{group}}$
(i.e. distribution under H_0)

(Parametric bootstrapping)

Plan

$$H_0: \sigma_u^2 = 0$$

$$H_A: \sigma_u^2 > 0$$

$$\text{test stat: } \hat{\rho}_{\text{group}}$$

p-value: requires us to approximate distribution of $\hat{\rho}_{\text{group}}$ under H_0

1) Pretend H_0 is true and fit reduced model
 $\text{Metabolic}_{ij} = \beta_0 + \varepsilon_{ij} \quad (\text{no } u_i)$

2) Pretend reduced model is correct and simulate data from reduced model

3) Fit full model (with random effect) on simulated data and calculate $\hat{\rho}_{\text{group}}$

4) Repeat steps 2 & 3 many times
(e.g. w/ a for loop)

Class activity

Work on the class activity. Submit your HTML on canvas at the end of class.